Instructions

Part 1: Upgrade an IOS Image on a Cisco Device

Step 1: Upgrade an IOS image on a router.

a. Access the TFTP server and enable the TFTP service.

b. Note the IOS image files that are available on the TFTP server.

Question:

Which IOS images stored on the server are compatible with a 1941 router?

Type your answers here.

c1900-universalk9-mz.SPA.151-4.M4.bin and c1900-universalk9-mz.SPA.155-3.M4a.bin

Open configuration window

c. From R2, issue the show flash: command and record the available flash memory.

Type your answers here.

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d. Copy the CISCO1941/K9 IOS version 15.5 image for the 1941 router from the TFTP Server to R2.

Note: In an actual network, if there is more than one interface active on the router, you may need to enter the ip tftp source interface command to specify which interface should be used to contact the TFTP server. This command is not supported in PT 7.2 and older versions and is not necessary to complete this activity.

```
R2# copy tftp: flash:
Address or name of remote host []? 192.168.2.254
Source filename []? c1900-universalk9-mz.SPA.155-3.M4a.bin
Destination filename [c1900-universalk9-mz.SPA.155-3.M4a.bin]?
Accessing tftp://192.168.2.254/c1900-universalk9-mz.SPA.155-3.M4a.bin....
Loading c1900-universalk9-mz.SPA.155-3.M4a.bin from 192.168.2.254:
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
[OK - 33591768 bytes]
```

```
33591768 bytes copied in 4.099 secs (860453 bytes/sec)
```

e. Verify that the IOS image has been copied to flash.

Question:
How many IOS images are located in flash?
Type your answers here.
2
f. Use the boot system command to load the version 15.5 IPBase image on the next reload.

```
R2(config)# boot system flash c1900-universalk9-mz.SPA.155-3.M4a.bin
```

g. Save the configuration and reload R2.
h. Use the show version command to verify the upgraded IOS image is loaded after R2 reboots.

```
R2# show version
Cisco IOS Software, C1900 Software (C1900-UNIVERSALK9-M), Version 15.5(3)M4a,
RELEASE
SOFTWARE (fc1)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2016 by Cisco Systems, Inc.
Compiled Thu 06-Oct-16 13:56 by mnguyen
ROM: System Bootstrap, Version 15.0(1r)M9, RELEASE SOFTWARE (fc1)
R2 uptime is 21 seconds
System returned to ROM by power-on
System image file is "flash0:c1900-universalk9-mz.SPA.155-3.M4a.bin"
---- output omitted -----
```

Close configuration window
Step 2: Upgrade an IOS image on a switch.
Open configuration window

a. Access the TFTP server and copy the c2960-lanbasek9-mz.150-2.SE4.bin image to S1.

```
S1# copy tftp: flash:
```

b. Use the boot system command to configure the switch to load the new IOS image on boot.
c. Reload S1 and verify the new image has been loaded into memory.
d. Close the TFTP configuration window if it is still open.
Close configuration window
Part 2: Backup an IOS Image to a TFTP Server
Open configuration window
a. On R1, display the contents of flash and record the IOS image.
Type your answers here.
isr4300-universalk9.03.16.05.S.155-3.S5-ext.SPA.bin

```
R1# show flash:
```

b. Use the copy command to back up the IOS image in flash memory on R1 to a TFTP server. Note: The isr4300 image is considerably larger than the c1900 image. It will take longer to transmit it to the TFTP server.

```
R1# copy flash: tftp:
Source filename []? isr4300-universalk9.03.16.05.S.155-3.S5-ext.SPA.bin
Address or name of remote host []? 192.168.2.254
Destination filename [isr4300-universalk9.03.16.05.S.155-3.S5-ext.SPA.bin]?
Writing
isr4300-universalk9.03.16.05.S.155-3.S5-ext.SPA.bin....!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
!!!!!!!!!!!!!! --- output omitted ----
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
!!!!!!!!!!!!!!!
[OK - 486899872 bytes]
486899872 bytes copied in 18.815 secs (83367 bytes/sec)
```

c. Access the TFTP server and verify that the IOS image has been copied to the TFTP server. Note: You may have to start and stop the TFTP service on the server so the file appears in the file listing.